

Cauchy's Rigidity Theorem

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IT College
Computer Science

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The back story

The setting...

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- It's the French revolution.



Viva la Revolution!

The back story

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- Legendre (*luh-zhahn-druh*) decides to translate the *Elements*.



Legendre ¹

¹The only know picture of Legendre. For centuries there was a different picture shown in books. The mistake wasn't discovered until 2005.

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- But he runs across a gap. . .
and hands it off to a 19 year old Cauchy.



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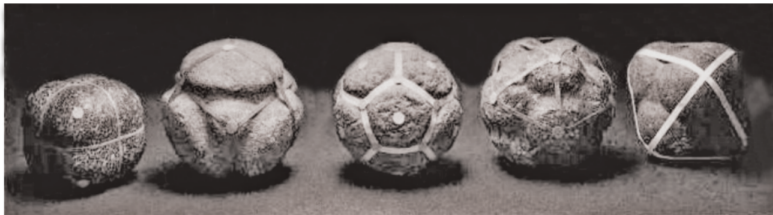
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A young, hip Cauchy

Polyhedron problem



So what's the problem?

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Let P and $P' \subset \mathbb{R}^3$ be two convex polytopes such that $\alpha(P) \simeq_{\pi} \alpha(P')$, and for every 2-dimensional face F , $F \cong F'$ (up to rigid motion by $O_3(\mathbb{R}) \ltimes \mathbb{R}^3$, where $F' = \pi(F)$). Then $P \cong P'$.

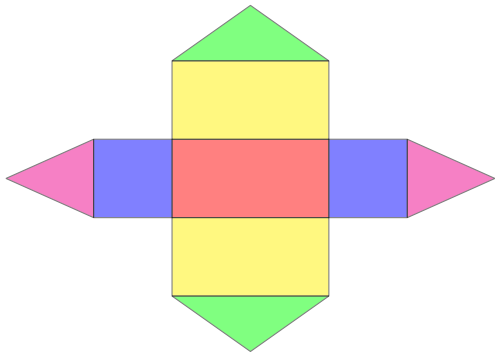
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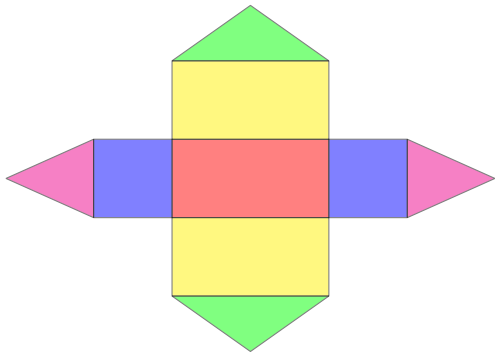
Given the faces and gluing instructions, there is only one way to construct a polyhedron.

The problem...

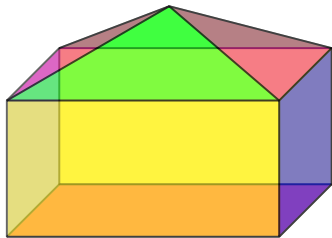


Unfolded polyhedron

The problem...

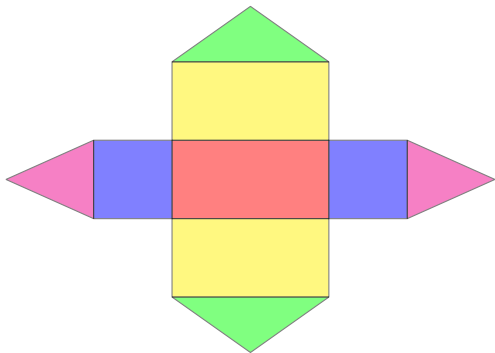


Unfolded polyhedron

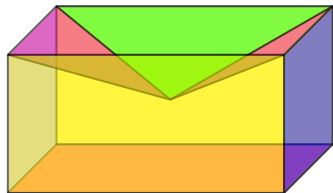


Convex polyhedron

The problem...

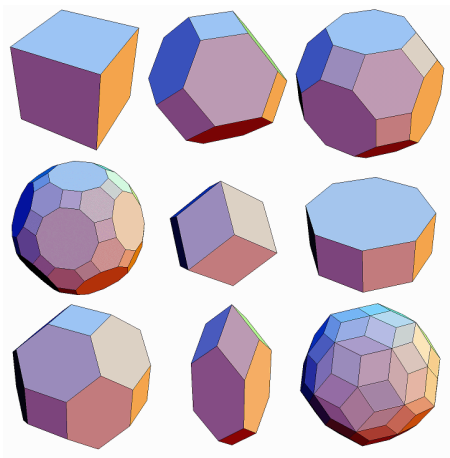


Unfolded polyhedron



Convexity matters!

True for any convex polyhedron



So what can change?

So what can change?

- Face angles fixed.

So what can change?

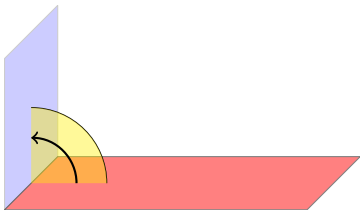
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So what can change?

- Face angles fixed.
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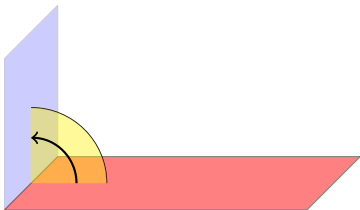
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- Only dihedral angles can change.



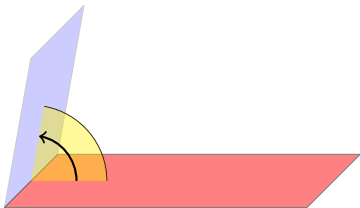
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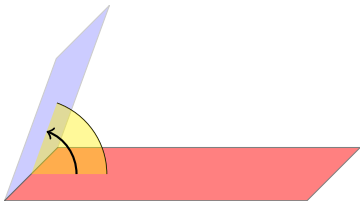
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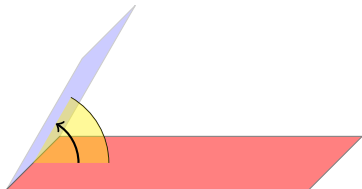
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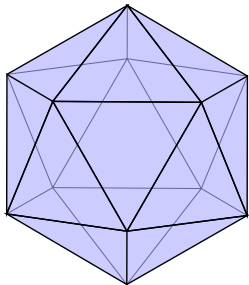
Plan of attack

If it could change, what would happen?

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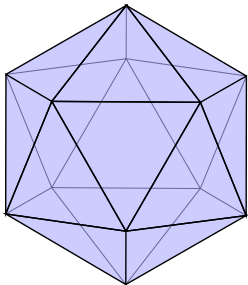
- What happens *globally*?



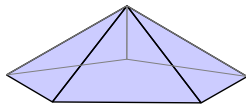
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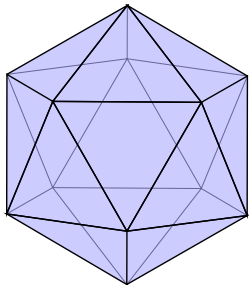
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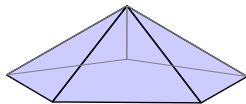
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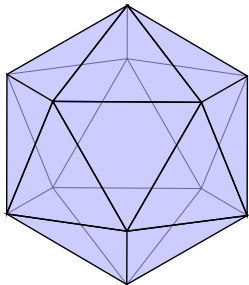


Global behavior $\Rightarrow$ upper limit on
hinge angle changes.

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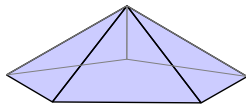
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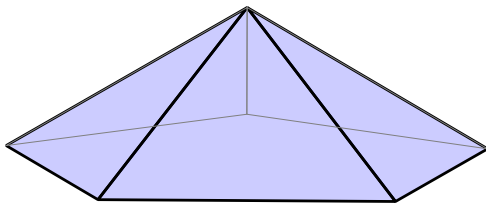


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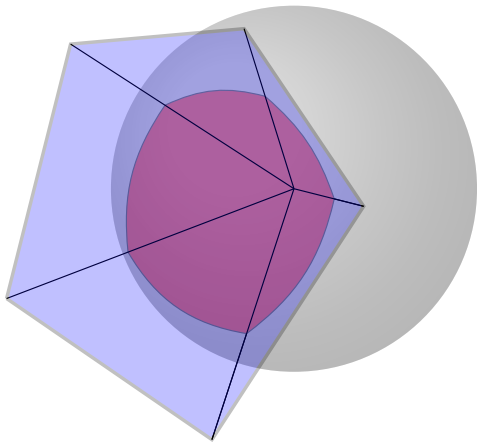
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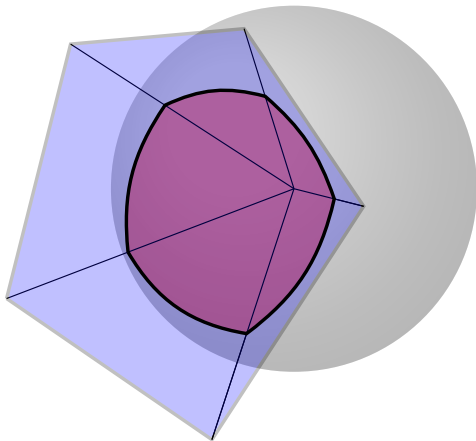
Local behavior $\Rightarrow$ lower limit on hinge angles change.



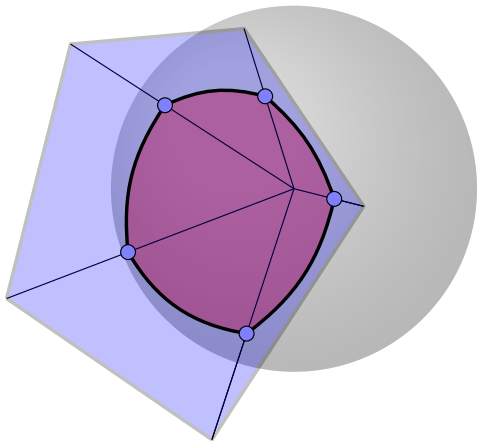
vertex



Stick the vertex into a sphere.

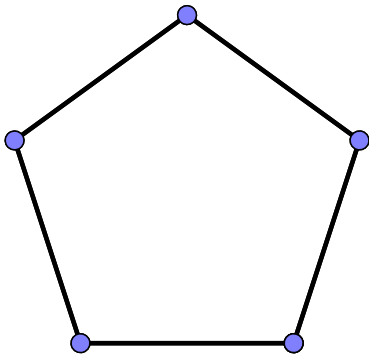


arcs = face angles



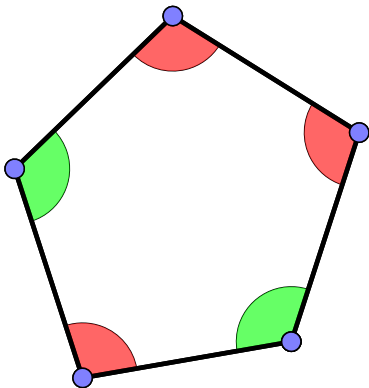
intersection is a polygon

Polygon angles = hinge angles



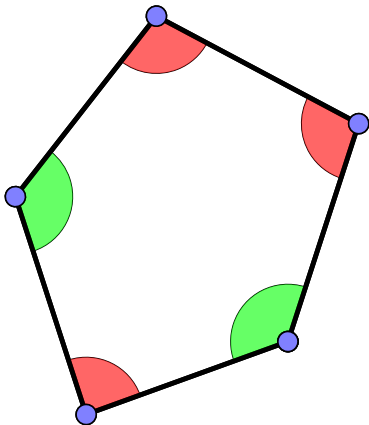
Polygon angles = hinge angles

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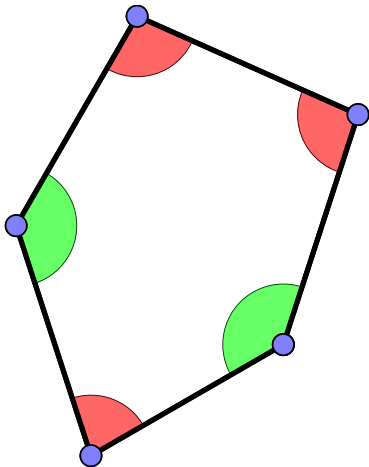
squishy

Polygon angles = hinge angles



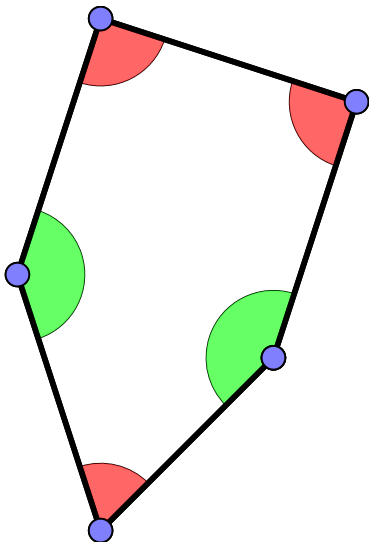
squishy

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squishier

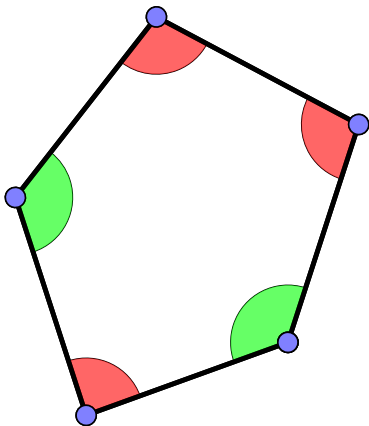
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Polygon angles = hinge angles

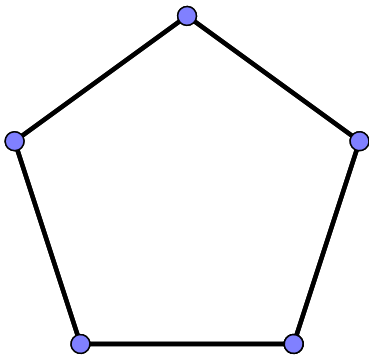
Count the *number* of changes. What's the least you can have?



Here we have 4 changes.

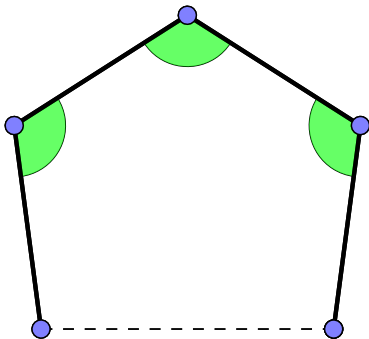
Polygon angles = hinge angles

What if we only have 2 changes?



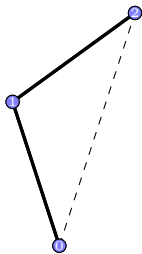
Polygon angles = hinge angles

Cauchy's arm lemma: There must be at least 4 changes.



snap!

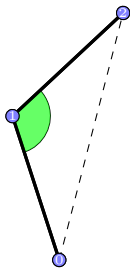
Cauchy's arm lemma



$$n = 3$$

Cauchy's arm lemma

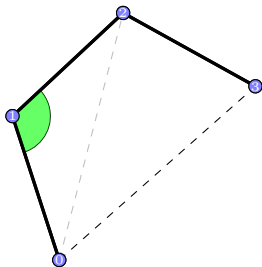
- Law of cosines $\Rightarrow n = 3$ case.



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Cauchy's arm lemma

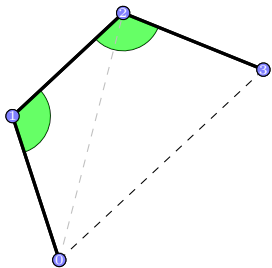
- Law of cosines $\Rightarrow n = 3$ case.
- Increase previous angles except v_2



$n = 4$

Cauchy's arm lemma

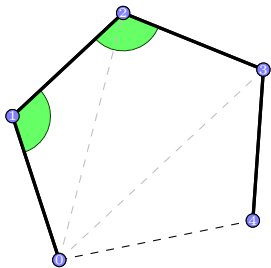
- Law of cosines $\Rightarrow n = 3$ case.
- Increase previous angles except v_2
- Increase v_2 ; reapply LOC.



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Cauchy's arm lemma

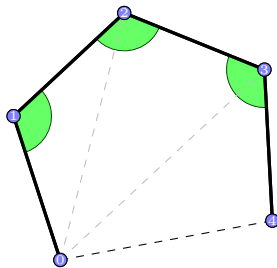
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Cauchy's arm lemma

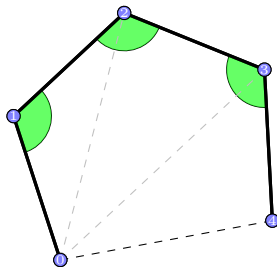
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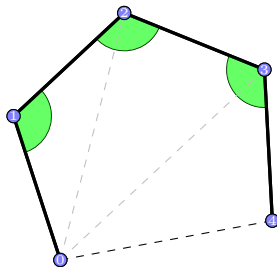
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$n = 1, 2, \dots$

Cauchy's arm lemma

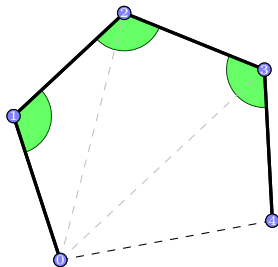
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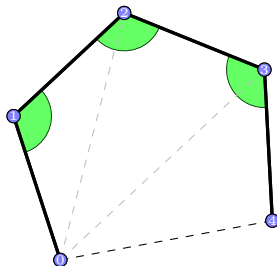
- Law of cosines $\Rightarrow n = 3$ case.
- If it's true for n



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Cauchy's arm lemma

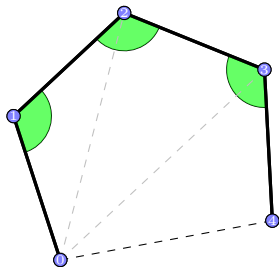
- Law of cosines $\Rightarrow n = 3$ case.
- If it's true for n
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Cauchy's arm lemma

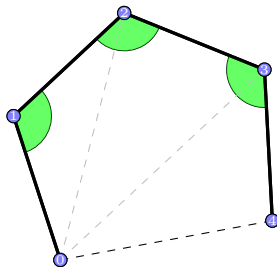
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- Reapply law of cosines.



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Cauchy's arm lemma

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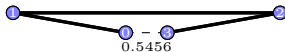


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Cauchy's arm lemma ☹

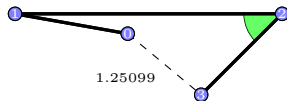
Cauchy's arm lemma ☹

- Too bad this is wrong!



Cauchy's arm lemma ☹

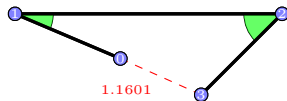
- Too bad this is wrong!
- the tragic loss of convexity.



Oops!

Cauchy's arm lemma ☹

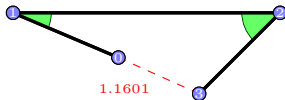
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- Unnoticed for +100 years.



Everybody makes mistakes. Even geniuses.

Cauchy's arm lemma ☹

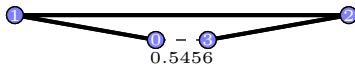
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- Unnoticed for +100 years.
- Geometry is tricky!



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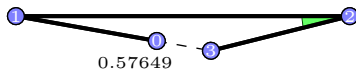
Cauchy's arm lemma ☺

- Fortunately, there is a slick fix.



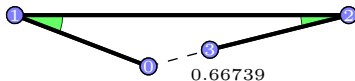
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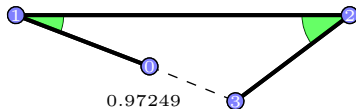
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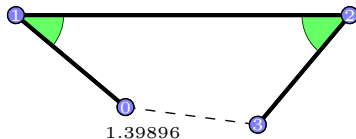
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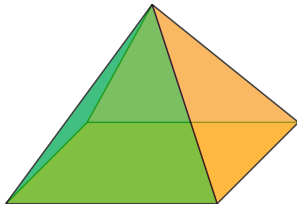


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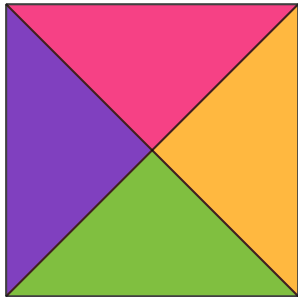
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Cauchy's arm lemma $\Rightarrow$ a *local* condition:

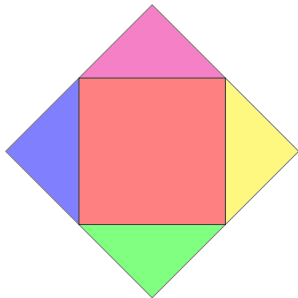
The number of changes (from $+$ to $-$) at a vertex is ≥ 4 .



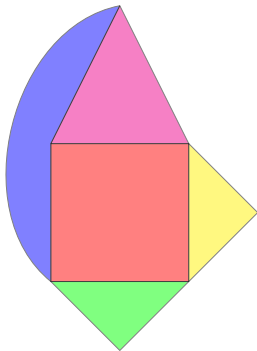
any polyhedron



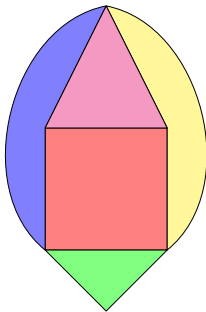
can be flattened out



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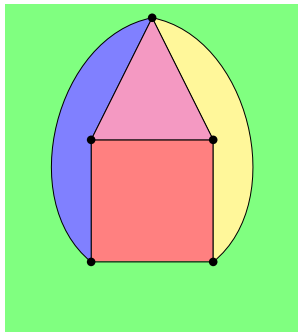


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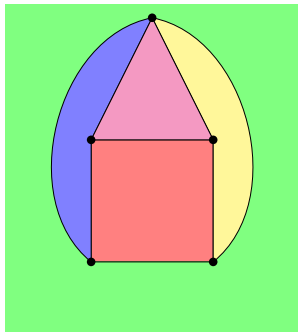
The geometry is gone.



Topological graph

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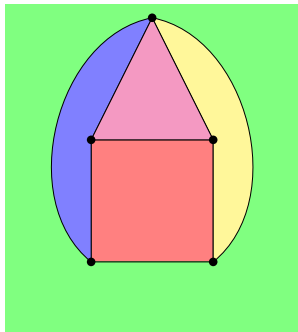
- vertices = $V = 5$



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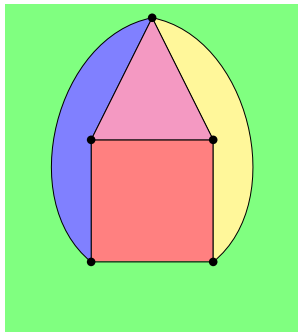
- vertices = $V = 5$
- edges = $E = 8$



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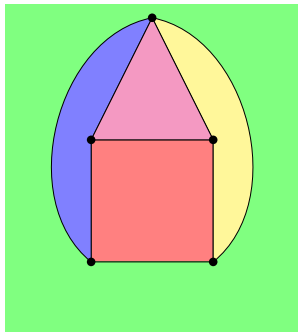
- vertices = $V = 5$
- edges = $E = 8$
- faces = $F = 5$



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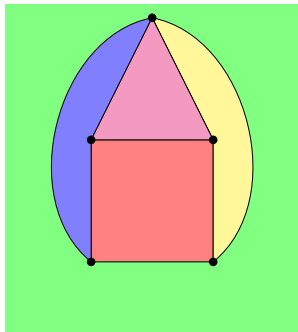
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- $5 - 8 + 5 = 2$



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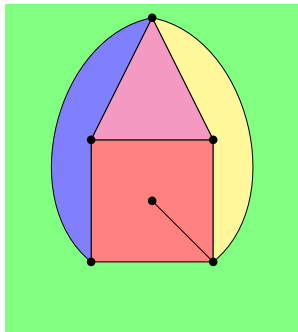
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Euler characteristic for a topological sphere:

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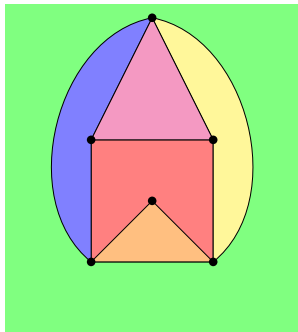
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- $5 - 8 + 5 = 2$



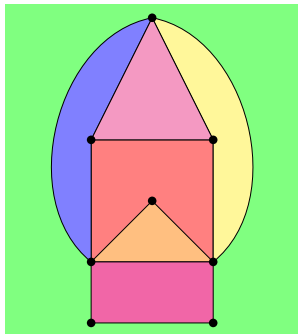
Topological graph

Euler characteristic for a topological sphere:

$$V - E + F = 2$$

The geometry is gone.

- vertices = $V = 5$
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- faces = $F = 5$
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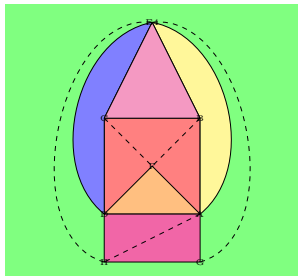
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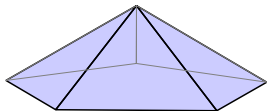


Topological graph

Euler characteristic for a topological sphere:

$$V - E + F = 2$$

M is the number of changes.



Local $M \geq 4V$

M is the number of changes.

Local $M \geq 4V$

- $F = \text{triangles} + \text{squares} + \text{pentagons} + \dots$

M is the number of changes.

Local $M \geq 4V$

- $F = f_3 + f_4 + f_5 \dots$

M is the number of changes.

Local $M \geq 4V$

- $F = f_3 + f_4 + f_5 \dots$
- $2 \cdot E = 3f_3 + 4f_4 + 5f_5 \dots$

M is the number of changes.

Local $M \geq 4V$

- $F = f_3 + f_4 + f_5 \dots$
- $2 \cdot E = 3f_3 + 4f_4 + 5f_5 \dots$
- $V - E + F = 2$

M is the number of changes.

Local $M \geq 4V$

- $4 \cdot F = 4f_3 + 4f_4 + 4f_5 \dots$
- $4 \cdot E = 6f_3 + 8f_4 + 10f_5 \dots$
- $V - E + F = 2$

M is the number of changes.

Local $M \geq 4V$

- $4 \cdot F = 4f_3 + 4f_4 + 4f_5 \dots$
- $4 \cdot E = 6f_3 + 8f_4 + 10f_5 \dots$
- $V - E + F = 2$
 $\Rightarrow 4V - 4E + 4F = 8$

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- $V - E + F = 2$
 $\Rightarrow 4V - 4E + 4F = 8$
 $\Rightarrow 4V - 8 = 4E - 4F$

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Local $M \geq 4V$

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 $\Rightarrow 4V - 4E + 4F = 8$
 $\Rightarrow 4V - 8 = 4E - 4F$
 $\Rightarrow 4V - 8 = 2f_3 + 4f_4 + 6f_5 \dots$

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Local $M \geq 4V$

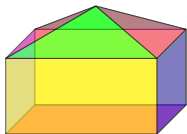
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- $V - E + F = 2$
 $\Rightarrow 4V - 4E + 4F = 8$
 $\Rightarrow 4V - 8 = 4E - 4F$
 $\Rightarrow 4V - 8 = 2f_3 + 4f_4 + 6f_5 \dots$

$$M > 4V - 8 = \\ 2f_3 + 4f_4 + 6f_5 + 8f_6 \dots$$

M is the number of changes.

Global

- Trick is to count M on faces.



Local $M \geq 4V$

- $4 \cdot F = 4f_3 + 4f_4 + 4f_5 \dots$
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- $V - E + F = 2$
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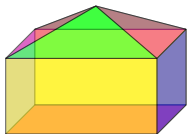
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Triangles: $2f_3$



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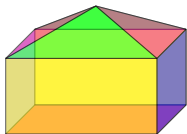
M is the number of changes.

Global

- Trick is to count M on faces.

Triangles: $2f_3$

Squares: $4f_4$



Local $M \geq 4V$

- $4 \cdot F = 4f_3 + 4f_4 + 4f_5 \dots$
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- $V - E + F = 2$
 $\Rightarrow 4V - 4E + 4F = 8$
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$$M > 4V - 8 = 2f_3 + 4f_4 + 6f_5 + 8f_6 \dots$$

Global vs. Local

M is the number of changes.

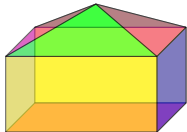
Global

- Trick is to count M on faces.

Triangles: $2f_3$

Squares: $4f_4$

Pentagons: $4f_5$



$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

Local $M \geq 4V$

- $4 \cdot F = 4f_3 + 4f_4 + 4f_5 \dots$
- $4 \cdot E = 6f_3 + 8f_4 + 10f_5 \dots$
- $V - E + F = 2$
 $\Rightarrow 4V - 4E + 4F = 8$
 $\Rightarrow 4V - 8 = 4E - 4F$
 $\Rightarrow 4V - 8 = 2f_3 + 4f_4 + 6f_5 \dots$

$$M > 4V - 8 = \\ 2f_3 + 4f_4 + 6f_5 + 8f_6 \dots$$

Global vs. Local: The finale

Global

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

Local $M > 4V$

$$M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Global vs. Local: The finale

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$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

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$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

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$$M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Assume two different polyhedrons exist. . .

Global vs. Local: The finale

Global

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

Local $M > 4V$

$$M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Assume two different polyhedrons exist. . .

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Global vs. Local: The finale

Global

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

Local $M > 4V$

$$M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Assume two different polyhedrons exist. . .

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

- But what if all hinge angles at k vertices don't change?

Global vs. Local: The finale

Global

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M$$

Local $M > 4V$

$$M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

Assume two different polyhedrons exist. . .

$$2f_3 + 4f_4 + 4f_5 + 6f_6 \cdots \geq M > 2f_3 + 4f_4 + 6f_5 + 8f_6 \cdots$$

- But what if all hinge angles at k vertices don't change?

Remove those vertices $\Rightarrow V - E + F = 2 + k$, which helps our case above.

200 years of relevance

- Non-convex polyhedron aren't necessarily rigid.
- First example was found in 1977 (Connelly).
- In 1997 it was shown that the volume is constant! (Connelly, Sabitov, and Walz)
- Many open questions about polyhedron remain.

Cauchy's Arm Lemma on a Growing Sphere

Zachary Abel* David Charlton† Sébastien Collette‡ Erik D. Demaine§¶
Martin L. Demaine§ Stefan Langerman|| Joseph O'Rourke** Val Pinciu††
Godfried Toussaint‡‡

November 2, 2018

Abstract

We propose a variant of Cauchy's Lemma, proving that when a convex chain on one sphere is redrawn (with the same lengths and angles) on a larger sphere, the distance between its endpoints increases. The main focus of this work is a comparison of three alternate proofs, to show the links between Toponogov's Comparison Theorem, Legendre's Theorem and Cauchy's Arm Lemma.

1 Introduction

A chain is composed of a sequence of points $v_0, \dots, v_n$, where each consecutive pair is connected by edges $e_i = v_{i-1}v_i$, and the angle at vertex v_i between e_i and e_{i+1} is θ_i . A chain is convex if the corresponding polygon (that is, if we add the edge v_0v_n to close the chain) is convex. These are well defined in the plane as well as on the sphere. In the case of the sphere, the edges are arcs of great circles. Cauchy's lemma can be expressed in the following way (although this was not its original statement):

A CHARACTERIZATION OF CONVEX HYPERBOLIC POLYHEDRA AND OF CONVEX POLYHEDRA INSCRIBED IN THE SPHERE

CRAIG D. HODGSON, IGOR RIVIN, AND WARREN D. SMITH

ABSTRACT. We describe a characterization of convex polyhedra in $\mathbf{H}^3$ in terms of their dihedral angles, developed by Rivin. We also describe some geometric and combinatorial consequences of that theory. One of these consequences is a combinatorial characterization of convex polyhedra in $\mathbf{E}^3$ all of whose vertices lie on the unit sphere. That resolves a problem posed by Jakob Steiner in 1832.

In 1832, Jakob Steiner in his book [23] asked the following question:

In which cases does a convex polyhedron have a (combinatorial) equivalent which is inscribed in, or circumscribed about, a sphere?

This was the 77th of a list of 85 open problems posed by Steiner, of which only numbers 70, 76, and 77 were still open as of last year. Apparently René Descartes was also interested in the problem (see [16]).

RIGIDITY OF CIRCLE POLYHEDRA IN THE 2-SPHERE AND OF HYPERIDEAL POLYHEDRA IN HYPERBOLIC 3-SPACE

JOHN C. BOWERS, PHILIP L. BOWERS, AND KEVIN PRATT

ABSTRACT. We generalize Cauchy's celebrated theorem on the global rigidity of convex polyhedra in Euclidean 3-space $\mathbb{E}^3$ to the context of circle polyhedra in the 2-sphere $\mathbb{S}^2$. We prove that any two convex and proper non-unitary c -polyhedra with Möbius-congruent faces that are consistently oriented are Möbius-congruent. Our result implies the global rigidity of convex inversive distance circle packings in the Riemann sphere as well as that of certain hyperideal hyperbolic polyhedra in $\mathbb{H}^3$.

INTRODUCTION

Start with an abstract oriented triangulation of the 2-sphere $\mathbb{S}^2$ in which each edge is labeled with a non-negative real number. A *circle realization* of this triangulation is a drawing of circles on the sphere, one circle for each vertex, that respects orientation, and with the inversive distance between adjacent circles equal to the label for that edge. When connecting adjacent circle centers by geodesic arcs results in a geodesic triangulation of

Unfolding Restricted Convex Caps

Joseph O'Rourke*

November 3, 2018

Abstract

This paper details an algorithm for unfolding a class of convex polyhedra, where each polyhedron in the class consists of a convex cap over a rectangular base, with several restrictions: the cap's faces are quadrilaterals, with vertices over an underlying integer lattice, and such that the cap convexity is “radially monotone,” a type of smoothness constraint. Extensions of Cauchy's arm lemma are used in the proof of non-overlap.

1 Introduction

Few classes of convex polyhedra are known to be *edge-unfoldable*: unfolded by cutting along edges of the polyhedron and flattening into the plane to a single piece without overlap. Among the classes known to be are: pyramids, prismoids, and “domes.” See [DO07, Chap. 22] for background on this problem.

The purpose of this informal note is to introduce another narrow class of

200 years of relevance



Paul Erdős has a theory that God has a book containing all the theorems of mathematics with their absolutely most beautiful proofs, and when he wants to express particular appreciation of a proof he exclaims, "This is from the book!"

— *Ross Honsberger* —

AZ QUOTES

200 years of relevance



Things people want to talk about

- Machine learning and AI algorithms, e.g. how do advertisers know what to show me? How do deep fakes work? How can AI help find struggling students? How can AI help make better course materials? Monstrous moonshine?
- Formal methods, some of my work with the NSA
- Details about our 5-12 courses to those outside Teacher's College
- Cauchy's thm, circle packing, Riemann's hypothesis, tiling, $P \neq NP(?)$
- Base number systems
- "A framework for understanding student's ways of knowing mathematics, and how it helps teaching them"
- Confocal microscopy in biomedical engineering or present a published paper from a math-related journal
- Strong differentiation in industry between engineers with strong math skills and engineers without. (Hint: more math is better).

Things people want to hear about

- Other ideas on teaching methods from other groups.
- Demonstration of a math subject matter such as Geometry, Trig, Probability.
- How to develop better math skills in students
- No preference
- Anything!
- Any and everything math related! I am in Elem Math Methods, but interested in all aspects of math.
- All of the things you described would be great.
- How folks are staying involved in mathematics, things that are working with your students, and good math reads (books, articles, blogs, etc.)
- Anything math or CI related is fine.
- Anything my peers would like to share

Things people want to hear about

- I would like to know more about math courses of study outside the TC.
- Previous and ongoing research interests, WGU-related research, online math teaching strategies
- Math and how to support students
- current events, challenges, impacts, solutions, etc.
- Maths
- Whatever topics other mathies feel like sharing
- CI strategies both personal and technological, AND 'fun' slightly-advanced math
- Current math news, neat facts, "on this day" in math history, teaching strategies, research interests; journal club - present published research from a math-related journal (not necessarily our own work) and allow discussion time

That is all